

Problem 1:

Prove, using induction, that any wff, constructed from $\{\wedge, \vee, \top, \perp\}$ using parentheses as usual, is monotonic.

Proof.

Base Case:

- $\top$. Then $\top(\top) = \top$ and $\top(\perp) = \top$ is monotonic trivially.
- $\perp$. Then $\perp(\top) = \perp$ and $\perp(\perp) = \perp$ is again monotonic trivially.
- p a variable. Then $p(\top) = \top$ and $p(\perp) = \perp$ is monotonic since changing a false to a true changes the output from a false to a true.

Inductive Step:

Assume that φ and ψ are constructed from $\{\wedge, \vee, \top, \perp\}$ and are monotonic.

If φ and ψ have no inputs then let their fixed values be $a_\varphi, b_\psi \in \{\text{False}, \text{True}\}$.

Then for every truth assignment v, w we have $\varphi * \psi(v) = a_\varphi * b_\psi = \varphi * \psi(w)$ for $*$ $\in \{\wedge, \vee\}$

If one of them has no inputs. WLOG let φ have no inputs.

If $a_\varphi = \text{True}$ then for all assignments v, w $\varphi \vee \psi(v) = \text{True} = \varphi \vee \psi(w)$, and $\varphi \wedge \psi = \psi$, so these are both monotonic.

If $a_\varphi = \text{False}$ then for all assignments $\varphi \wedge \psi(v) = \text{False} = \varphi \wedge \psi(w)$, and $\varphi \vee \psi = \psi$, so these are again both monotonic.

If φ and ψ have at least one input.

Let v, w be truth assignments for the wffs $\varphi * \psi$ $*$ $\in \{\wedge, \vee\}$ that differ on exactly one variable p , with $v(p) = \text{False}$ and $w(p) = \text{True}$.

- Case $*$ $= \wedge$

If $\varphi(v) = \text{True}$ and $\psi(v) = \text{True}$ then by the Induction hypothesis we have that $\varphi(w) = \psi(w) = \text{True}$. Hence,

$$\varphi \wedge \psi(v) = \text{True} = \varphi \wedge \psi(w)$$

Otherwise, $\varphi \wedge \psi(v) = \text{False}$, so certainly either $\varphi \wedge \psi(w)$ is true or false in both cases it shows $\varphi \wedge \psi$ is monotonic.

- Case $*$ $= \vee$

If $\varphi(v) = \text{True}$ (or symmetrically $\psi(v) = \text{True}$), then $\varphi(w) = \text{True}$ since it is monotonic, so

$$\varphi \vee \psi(v) = \text{True} = \varphi \vee \psi(w)$$

Otherwise, $\varphi(v) = \psi(v) = \text{False}$, so $\varphi \vee \psi(v) = \text{False}$, so again either $\varphi \vee \psi(w)$ is true or false, but in both cases it shows $\varphi \vee \psi$ is monotonic.

Finally, if v, w differ s.t. $v(p_i) = \text{False}$ and $w(p_i) = \text{True}$ for $i \in \{1, \dots, n\}$ then flip those variables one at a time from false to true applying the single flip result at each step.

Therefore, $\varphi * \psi$ are monotonic.

Therefore, by induction wffs made from $\{\wedge, \vee, \top, \perp\}$ are monotonic. ■

Problem 2:

Prove that $\{\wedge, \vee, \top, \perp\}$ is complete for the **MONOTONIC** boolean functions, where here $\top$ and $\perp$ take no inputs.

Proof.

Fix variables. View $\{0, 1\}^n$ as a partially ordered set:

$$x \leq y \iff x_i \leq y_i \text{ for all } i$$

Consider a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ that is monotonic.

Let

$$T = \{x \in \{0, 1\}^n \mid f(x) = 1\}$$

be the true set of f .

Let $M = \{a \in T \mid \nexists b \in T \text{ s.t. } b < a\}$.

Since $\{0, 1\}^n$ is finite it has minimal elements and if $T \neq \emptyset$ then $M \neq \emptyset$.

For each $a = (a_1, \dots, a_n) \in \{0, 1\}^n$, define

$$C_a = \bigwedge_{i \in A} p_i$$

where $A = \{i \in \{1, \dots, n\} \mid a_i = 1\}$

Define

$$\varphi_f = \bigvee_{a \in M} C_a$$

If $M = \emptyset$ then replace with $\perp$.

I claim that φ_f is equivalent to f .

For $x \in \{0, 1\}^n$

If $\varphi_f(x) = 1$, then $C_a(x) = 1$ for some $a \in M$. Thus, $a \leq x$. Since $a \in T$ and f is monotonic this implies that $x \in T$ thus $f(x) = 1$.

If $f(x) = 1$, then $x \in T$. Then consider the finite nonempty set $S = \{y \in T \mid y \leq x\}$.

There is a minimal element a of S . Then $a \in M$ and $a \leq x$, so $C_a(x) = 1$, hence $\varphi_f(x) = 1$.

Thus, f, φ_f agree on all inputs.

Edge Cases:

If $f(x) = 0$ for all inputs, then $T = \emptyset$ and $\varphi_f = \perp$.

If $f(x) = 1$ for all inputs, then $T = \{0, 1\}^n$ and $M = \{(0, \dots, 0)\}$. Thus, $C_{(0, \dots, 0)}$ is the empty conjunction $\top$, so $\varphi_f = \top$. ■

Problem 3:

Prove, using induction, that wffs built from the **binary (2-input)** operators $\wedge, \vee$ and sentences can not express any of $\top, \perp, \neg$

Claim

For every wff φ built from sentence letters using only $\wedge$ and $\vee$ (with at least one sentence symbol):

- If v assigns False to every letter that occurs in φ , then $\varphi(v) = \text{False}$.
- If v assigns True to every letter that occurs in φ then $\varphi(v) = \text{True}$.

Proof.

Base Case:

If $v(p) = \text{False}$ then $p(v) = \text{false}$, and if $v(p) = \text{true}$ then $p(v) = \text{false}$.

Inductive Step:

Assume that φ, ψ are wffs built from sentences letters using only $\wedge$ and $\vee$, and the claim holds.

Let v be a truth assignment that sets all sentence symbols in $\varphi * \psi$ false, where $*$ $\in \{\wedge, \vee\}$. Since v sets all sentence symbols false this implies that every sentence symbol of φ, ψ is false. Thus, by the induction hypothesis, $\varphi(v_1) = \text{False}$, and $\psi(v_2) = \text{False}$ where v_1, v_2 are truth assignments that are all false.

Therefore, $(\varphi * \psi)(v) = \varphi(v_1) * \psi(v_2) = \text{False} * \text{False}$. If $*$ $= \wedge$ then $\text{False} \wedge \text{False} = \text{False}$, and if $*$ $= \vee$ then $\text{False} \vee \text{False} = \text{False}$.

There is a similar argument for all the sentence symbols being true.

Thus, the claim is true. ■

Proof.

Do the proof in problem 1 but just remove $\top$ and $\perp$ to prove that all sentence symbols made from $\wedge$ and $\vee$ are monotonic. This implies that any wff made only from $\wedge$ and $\vee$ cannot express $\neg$ since it is not monotonic we cannot express $\neg$ using just $\wedge$ and $\vee$.

If a wff contains at least one sentence symbol, then by the claim above it is False under the all-False assignment to its symbols, and also it is True under the all-True assignment. Hence, its truth value is non-constant. Therefore, $\top$ and $\perp$ cannot be expressed from just $\wedge$ and $\vee$. ■